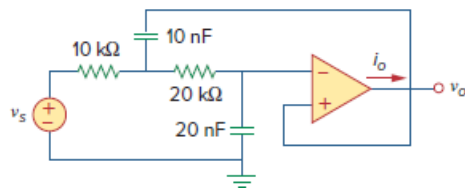


Problem

Find v_o and i_o in the op amp circuit of Fig. 10.32. Let $v_s = 12 \cos 5000t$ V.

Figure 10.32:



Step-by-step solution

Step 1 of 9

Refer to Figure 10.32 in the textbook.

Consider the source voltage.

$$v_s = 12 \cos 5000t \text{ V}.$$

Here,

$$\omega = 5000 \text{ rad/s}$$

Convert the time domain voltage to phasor domain.

$$12 \cos 5000t \text{ V} \Rightarrow 12 \angle 0^\circ \text{ V}$$

$$v_s = 12 \angle 0^\circ \text{ V}$$

Step 2 of 9

Determine the capacitive reactance for $C = 10 \text{ nF}$.

$$\begin{aligned} X_c(C = 10 \text{ nF}) &= \frac{1}{j\omega C} \\ &= \frac{1}{j(5000 \text{ rad/s})(10 \text{ nF})} \\ &= \frac{1}{j0.00005} \Omega \\ &= -j20 \text{ k}\Omega \end{aligned}$$

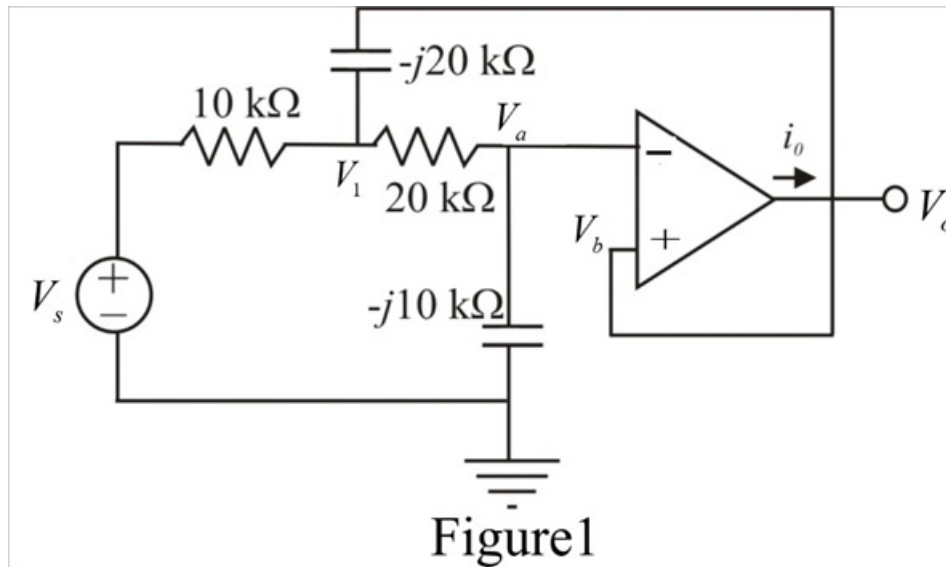
Step 3 of 9

Determine the capacitive reactance for $C = 20 \text{ nF}$.

$$\begin{aligned} X_c(C = 20 \text{ nF}) &= \frac{1}{j\omega C} \\ &= \frac{1}{j(5000 \text{ rad/s})(20 \text{ nF})} \\ &= \frac{1}{j0.0001} \Omega \\ &= -j10 \text{ k}\Omega \end{aligned}$$

Step 4 of 9

Draw the equivalent frequency domain circuit as shown in Figure 1



Step 5 of 9

From Figure 1, it is cleared that V_o is feedback at the non-inverting (+ve) end of op-amp.

Thus, voltage at b is equal to the output voltage i.e. $V_b = V_o$.

Also,

It is known that voltage at a is equal to voltage at b, i.e. $V_a = V_b$ due to virtual grounding.

Therefore,

It can be shown that voltage V_a is equal to V_o .

$$V_a = V_o \quad (\because V_a = V_b)$$

Step 6 of 9

Write the Kirchhoff's current law equations using Figure 1.

$$\frac{V_s - V_1}{10 \text{ k}} = \frac{V_1 - V_o}{-j20 \text{ k}} + \frac{V_1 - V_a}{20 \text{ k}}$$

Substitute V_o for V_a .

$$\frac{V_s - V_1}{10} = \frac{V_1 - V_o}{-j20} + \frac{V_1 - V_o}{20}$$

$$\frac{V_s}{10} + \frac{V_o}{20} - \frac{V_o}{j20} = \frac{V_1}{20} + \frac{V_1}{10} - \frac{V_1}{j20}$$

$$\frac{V_s}{10} + V_o \left(\frac{1}{20} - \frac{1}{j20} \right) = V_1 \left(\frac{1}{20} + \frac{1}{10} - \frac{1}{j20} \right)$$

$$0.1V_s + V_o (0.05 + j0.05) = V_1 (0.05 + 0.1 + j0.05)$$

On further simplification.

$$0.1V_s + V_o (0.05 + j0.05) = V_1 (0.15 + j0.05)$$

$$V_s + V_o (0.5 + j0.5) = V_1 (1.5 + j0.5)$$

$$V_s = -V_o (0.5 + j0.5) + V_1 (1.5 + j0.5)$$

$$V_s = -V_o (0.5 + j0.5) + V_1 (1.5 + j0.5) \dots\dots (1)$$

Step 7 of 9

Apply KCL at a .

$$\frac{V_1 - V_a}{20} = \frac{V_a - 0}{-j10}$$

Substitute V_0 for V_a .

$$\frac{V_1 - V_o}{20} = \frac{V_o - 0}{-j10}$$

$$-j10V_1 = (20 - j10)V_o$$

$$V_1 = V_o(1 + j2) \dots\dots (2)$$

Substitute $V_o(1 + j2)$ for V_1 in equation (1).

$$V_s = -V_o(0.5 + j0.5) + V_1(1.5 + j0.5)$$

$$V_s = -V_o(0.5 + j0.5) + V_o(1 + j2)(1.5 + j0.5)$$

$$V_s = V_o[-(0.5 + j0.5) + (1 + j2)(1.5 + j0.5)]$$

$$V_s = V_o[-0.5 - j0.5 + 0.5 + j3.5]$$

$$V_s = V_o[j3]$$

Step 8 of 9

Substitute $12\angle 0^\circ$ V for V_s .

$$12\angle 0^\circ = V_o[j3]$$

$$12\angle 0^\circ = V_o(3\angle 90^\circ)$$

$$\begin{aligned} V_o &= \frac{12\angle 0^\circ}{3\angle 90^\circ} \\ &= 4\angle -90^\circ \text{ V} \end{aligned}$$

Convert the phasor value of voltage V_o to time domain.

$$\begin{aligned} v_o &= 4 \cos(5000t - 90^\circ) \\ &= 4 \sin(5000t) \text{ V} \end{aligned}$$

Therefore, the value of voltage v_o is $\boxed{4 \sin 5000t \text{ V}}$.

Step 9 of 9

Calculate the value of current i_o .

$$\mathbf{I}_o = \frac{\mathbf{V}_o - \mathbf{V}_1}{-j20 \text{ k}}$$

$$\mathbf{I}_o = \frac{\mathbf{V}_o - \mathbf{V}_o(1+j2)}{-j20 \text{ k}} \quad (\because \mathbf{V}_1 = \mathbf{V}_o(1+j2))$$

$$\mathbf{I}_o = \frac{-2\mathbf{V}_o}{-20 \text{ k}}$$

Substitute $\mathbf{V}_o = 4\angle -90^\circ \text{ V}$.

$$\mathbf{I}_o = \frac{-2(4\angle -90^\circ)}{-20 \text{ k}}$$

$$= \frac{8\angle -90^\circ}{20 \text{ k}}$$

$$= 0.4 \times 10^{-3} \angle -90^\circ \text{ A}$$

$$= 400 \angle -90^\circ \mu\text{A}$$

Convert the value of current $\mathbf{I}_o$ to time domain.

$$i_o = 400 \cos(5000t - 90^\circ)$$

$$= 400 \sin(5000t) \mu\text{A}$$

Therefore, the required value of current i_o is $400 \sin 5000t \mu\text{A}$.